Algebra: The Distance Formula

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Name: ____________________________ Date: ______________ Score: ____ / 10

DIRECTIONS

Use the Distance Formula $d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$ to find the distance between each pair of points. Simplify all radicals.

1. Find d:

(0, 0) and (3, 4)

Answer: ______________________________

2. Find d:

(1, 2) and (4, 6)

Answer: ______________________________

3. Find d:

(−3, 0) and (0, 4)

Answer: ______________________________

4. Find d:

(2, −1) and (2, 5)

Answer: ______________________________

5. Find d:

(−1, −1) and (2, 3)

Answer: ______________________________

6. Find d:

(0, −3) and (4, 0)

Answer: ______________________________

7. Find d:

(−2, 3) and (4, −5)

Answer: ______________________________

8. Find d:

(3, −2) and (−1, 1)

Answer: ______________________________

9. Find d:

(−5, 2) and (3, −4)

Answer: ______________________________

10. Which is farther? A: (0,0)→(6,8) or B: (−3,1)→(3,−7)

Show work for both.

Answer: ______________________________

Based on the Numberbender lesson "ALGEBRA: Using the Distance Formula" • youtu.be/5JilGerPyW4

Answer Key & Solutions

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TEACHER NOTES

Items 1–6: results are perfect squares — no radical simplification needed. Items 7–8: results simplify cleanly. Items 9–10: compare or apply distances. Remind students to square both differences before adding under the radical.

1. Find d: (0, 0) and (3, 4)

Answer: 5

$\sqrt{(3^2+4^2)} = \sqrt{(9+16)} = \sqrt{25} = 5$.

2. Find d: (1, 2) and (4, 6)

Answer: 5

$\sqrt{(3^2+4^2)} = \sqrt{(9+16)} = \sqrt{25} = 5$.

3. Find d: (−3, 0) and (0, 4)

Answer: 5

$\sqrt{(3^2+4^2)} = \sqrt{(9+16)} = \sqrt{25} = 5$.

4. Find d: (2, −1) and (2, 5)

Answer: 6

$\sqrt{(0^2+6^2)} = \sqrt{36} = 6$.

5. Find d: (−1, −1) and (2, 3)

Answer: 5

$\sqrt{(3^2+4^2)} = \sqrt{(9+16)} = \sqrt{25} = 5$.

6. Find d: (0, −3) and (4, 0)

Answer: 5

$\sqrt{(4^2+3^2)} = \sqrt{(16+9)} = \sqrt{25} = 5$.

7. Find d: (−2, 3) and (4, −5)

Answer: 10

$\sqrt{(6^2+8^2)} = \sqrt{(36+64)} = \sqrt{100} = 10$.

8. Find d: (3, −2) and (−1, 1)

Answer: 5

$\sqrt{(4^2+3^2)} = \sqrt{(16+9)} = \sqrt{25} = 5$.

9. Find d: (−5, 2) and (3, −4)

Answer: 10

$\sqrt{(8^2+6^2)} = \sqrt{(64+36)} = \sqrt{100} = 10$.

10. Which is farther? A: (0,0)→(6,8) or B: (−3,1)→(3,−7) Show work for both.

Answer: Equal — both d = 10

A: $\sqrt{(36+64)}=10$; B: $\sqrt{(36+64)}=10$. Same distance.